



FOODPUZZLE: Developing Large Language Model Agents as Flavor Scientists

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Abstract

Flavor development in the food industry is increasingly challenged by the need for rapid innovation and precise flavor profile creation. Traditional flavor research methods typically rely on iterative, subjective testing, which lacks the efficiency and scalability required for modern demands. This paper presents three contributions to address these challenges. Firstly, we define a new problem domain for scientific agents in flavor science, conceptualized as the generation of hypotheses for flavor profile sourcing and understanding. By leveraging their capacity to identify relevant evidence and reason within large context spaces, language model-backed agents can perform the labor-intensive tasks of flavor sourcing and understanding with enhanced efficiency and precision. To facilitate research in this area, we introduce the FOODPUZZLE, a challenging benchmark consisting of 978 food items and 1,766 flavor molecule profiles. We propose a novel Scientific Agent approach, integrating in-context learning and retrieval augmented techniques to generate grounded hypotheses in the domain of food science. Experimental results indicate that our model significantly surpasses traditional methods in flavor profile prediction tasks, demonstrating its potential to transform flavor development practices.

CCS Concepts

• **Computing methodologies** → *Natural language processing*.

*Both authors contributed equally to this research.

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1 Introduction

The burgeoning demand for novel and appealing flavors in the food industry necessitates expedited innovation cycles without compromising on quality [12]. Traditionally, the process of ingredient sourcing and evaluation for flavor development has relied on manual assessment and human expertise. In addition to being labor-intensive, and subjective, this process also relies on time-consuming iterations, hindering the pace of flavor development [7, 12, 23]. Additionally, complex flavor interactions, together with the vast array of available ingredients, present formidable challenges for food scientists in optimizing flavor profiles [11].

Recent advancements in Large Language Models (LLMs) have revolutionized traditional scientific methodologies across a broad spectrum of disciplines. In fields ranging from biology and chemistry to physics and material sciences, researchers have employed LLM technologies to analyze complex datasets, uncover hidden patterns, and narrow down the search spaces of scientific problems [13, 14, 22, 29, 30]. This trend highlights the transformative potential of LLMs in reshaping traditional paradigms and catalyzing breakthroughs. This paper explores how the integration of LLMs can streamline and enhance the process of evaluating ingredient sources for their flavor-giving potential. However, the application of LLMs in ingredient sourcing and flavor formulation is hindered by

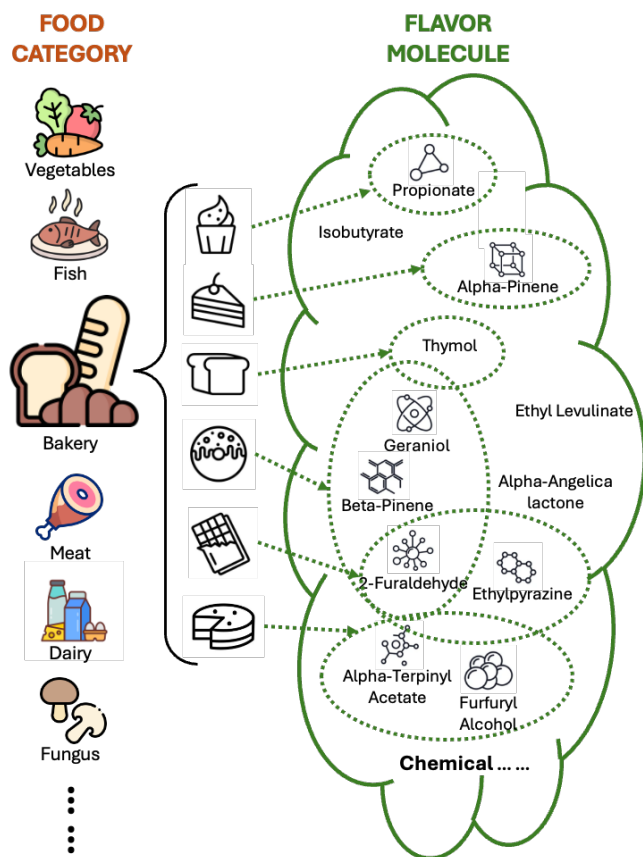


Figure 1: Flavor is determined by diverse flavor molecules. Sourcing and identifying these molecules from various foods is a time-consuming and resource-intensive task for food scientists. Understanding these connections is crucial in flavor science for developing and enhancing food products to ensure appealing taste experiences for consumers. In this work, we explore how LLMs can assist in this process.

a significant barrier: the absence of high-quality, domain-specific datasets. Most LLMs are trained on generic data from the Web, which often lacks the detailed, specialized knowledge required for accurate flavor profile understanding and prediction.

In this paper, we delineate three primary contributions that collectively advance the application of LLMs in the flavor science. Our *first* contribution is to formulate flavor profile sourcing and understanding as an LLM agent problem. Flavor sourcing and understanding are inherently complex tasks, involving an open-ended exploration of vast ingredient arrays to pinpoint components that deliver desired profiles. Scientific agents excel at navigating these complexities by leveraging their capacity to identify relevant evidence and reason within large context spaces effectively. When integrated with external knowledge sources, these agents can perform the labor-intensive tasks of flavor sourcing and understanding with enhanced efficiency and precision.

Our *second* contribution is the creation of the FOODPUZZLE dataset. To gather data for our study on flavor profile prediction and understanding, we turn to FlavorDB [8]. FlavorDB is a comprehensive database that includes detailed information on 25,595 flavor molecules, of which 1,766 are specifically noted for their presence in 978 natural ingredients. Figure 2 presents an overview of the structure of our constructed dataset. We propose two tasks that mirror real-world scientific study. The first task is Molecular Food Prediction (MFP), which challenges the model to predict possible food items based on a given set of flavor molecules. The second task is completing the molecular profile (MPC) which requires the identification of flavor molecules likely present in a specified food item.

As we realize the challenges of existing approaches for solving the FoodPuzzle task, our *third* contribution is the development of a comprehensive scientific agent method for proposing grounded hypotheses of flavor profile prediction tasks.¹ By integrating in-context learning (ICL) and the Retrieval-Augmented Generation (RAG) approach, our method enhances the model’s ability to produce accurate and reliable results while providing traceable justifications for its predictions, thus addressing key domain-specific knowledge and improving reliability in flavor development applications. Experimental results show that our proposed scientific agent method significantly outperforms baseline methods.

Our work represents a pilot study to design scientific agents for flavor profile sourcing tasks. We set a new benchmark for leveraging advanced AI technologies in the domain of flavor science. Our methodology provides clear, traceable insights into how flavor profiles are derived, thus facilitating greater acceptance and use of LLMs in practical food science applications.

2 Related Works

Retrieval-Augmented Generations. Retrieval-Augmented Language Models (RALMs) enhance the reasoning process by incorporating external knowledge sources, thus providing additional contextual information [2, 4, 15–17, 19, 37]. Early works such as Retrieval-Augmented Generation (RAG [20]) and RALM pre-training (REALM [10]) have demonstrated significant improvements in contextual understanding and question-answering by fetching relevant documents from large corpora. In contrast to conventional RAG approaches retrieving from a static index base, our method leverages scholarly articles from the internet and augments model reasoning with consolidated information about flavor science. This approach innovates upon conventional RAG systems and highlights the extensibility of our Flavor Scientific Agent.

LLMs for Scientific Research. The application of LLMs in scientific research has led to significant advancements across various disciplines. For example, domain-specific adaptations of BERT [5] for biology and biomedical domains have demonstrated significant performance improvements in tasks such as protein classification and text mining, underscoring the value of tailored pretraining for scientific corpora [1, 9, 18, 32, 34]. In the field of chemistry, LLMs have been tailored for tasks such as reaction prediction and

¹Our code and data is publicly available at https://tenghao Huang.github.io/FoodPuzzle_website/.

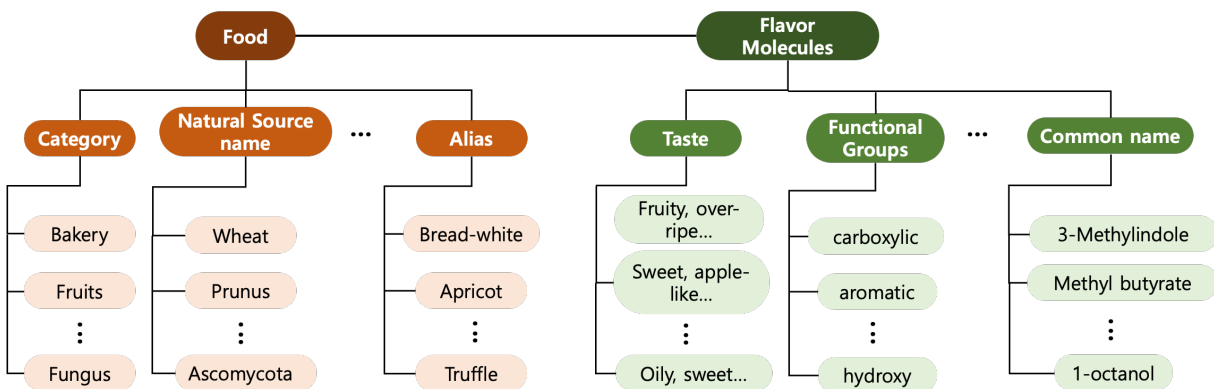


Figure 2: A high level overview of the FOODPUZZLE data hierarchy

molecule translation [6, 21, 27]. However, they face common challenges, including a dependency on large, high-quality scientific datasets for training, which are often difficult to collect, and a lack of interpretability in their predictions, making it challenging for chemists to trust and use the results effectively [3, 14, 31]. Our method leverages retrieval augmented techniques to provide in-context learning at inference time, outperforming training-time methods. Our approach not only streamlines the process by generating and selecting the best hypothesis from multiple candidates but also enhances interpretability by providing explicit evidence of the rationales behind the outputs.

3 Task and Data

This section delineates the tasks and datasets designed to evaluate the efficacy of LLMs in predicting food items based on their molecular flavor profiles. These tasks assess LLMs’ capabilities for contextually-enhanced reasoning within the domain of molecular data.

3.1 Task Definition

Molecular Food Prediction (MFP). We define the task as learning the mapping function h which takes a set of molecules $M = \{m_1, m_2, \dots, m_n\}$ as input and outputs a corresponding food source F . This can be expressed as:

$$F = h(M).$$

This formulation aims to predict food sources based solely on their molecular composition, without reliance on specific algorithms or systems.

Molecular Profile Completion (MPC). The primary objective of the Molecular Profile Completion (MPC) task is to identify the missing molecules needed to complete the molecular profile of a given food item. The function g receives the known food item F , a partial set of its molecules M_{partial} , and an integer n representing the expected number of missing molecules. It then outputs the set of missing molecules M_{missing} . This can be mathematically represented as:

$$M_{\text{missing}} = g(F, M_{\text{partial}}, n).$$

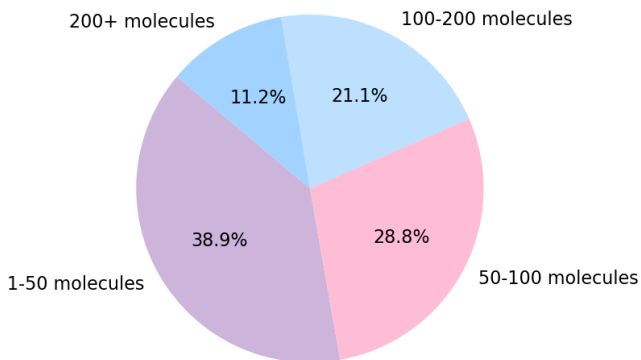


Figure 3: Distribution of the number of flavor molecules in the FOODPUZZLE dataset.

In this task, the integer n is provided as part of the input, indicating the number of molecules that are missing.

3.2 Constructing the FOODPUZZLE Dataset

Data Collection. To gather data for our study on flavor molecule prediction and analysis, we crawl through the extensive collection of food entries of the the Flavor DB ². From the collected data, we create three distinct stores of information. The first store contains profile information pertaining to each molecule (e.g., 2-Ethylpyrazine), detailing its properties (e.g., isotope atom count) and flavor characteristics (e.g., odor). The second store focuses on information related to each food item, categorizing them by type and listing associated molecular compositions. Finally, we construct an association matrix that mapped which foods were associated with which molecules, represent as pairs of food and molecule identifiers. This structured approach ensures that our dataset was both comprehensive and well-organized, facilitating subsequent analysis and modeling efforts. The resulting FOODPUZZLE dataset

²The address of FlavorDB endpoint - <https://cosylab.iitd.edu.in/flavordb>

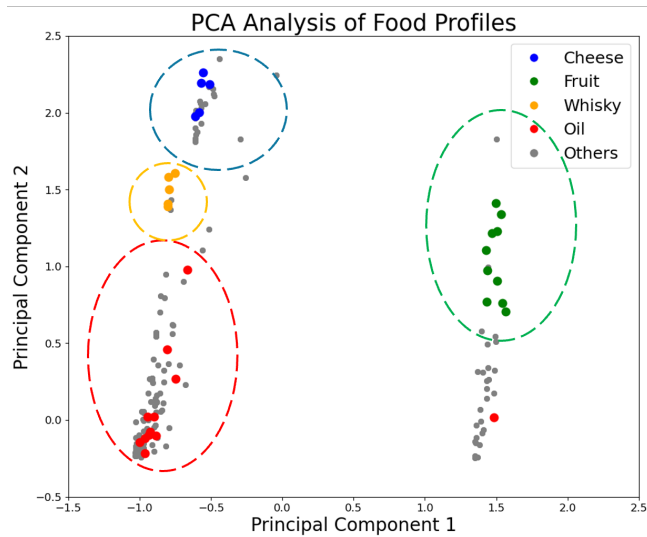


Figure 4: PCA visualization illustrating clustering of food entities based on molecular profiles.

maps 978 foods to 1766 flavor molecules. For the purpose of fine-tuning on the baseline, we split the dataset into train/dev/test sets by 80%/10%/10%.

Data Analysis. Our dataset includes a variety of food entities, each associated with a unique set of molecules. Figure 3 reveals that the majority of food entities are associated with a relatively small number of molecules, while a few entities have a significantly higher number of associated molecules. We employed feature encoding to represent each food item by its molecular composition, resulting in each item being characterized by a 1766-dimensional vector. Principal Component Analysis (PCA) of the encoded vectors, illustrated in Figure 4, reveals distinct clustering for categories such as cheese, whisky, oil, and fruit. These clusters indicate that food entities with analogous molecular compositions tend to aggregate, suggesting inherent patterns in flavor molecules that could potentially be predicted using machine learning techniques.

3.3 Evaluation Protocols

Evaluation of Molecular Food Prediction (MFP). The objective of MFP is to predict the category of a food item based on its molecular composition. There are 21 categories as shown in Table 1. To measure the correctness of predicted food category, we provide the groundtruth answer F^* , which is sourced from FlavorDB. At evaluation time our models predict a food category $\hat{F}$ but this is done as a free text prediction; no ontology is provided. We thus compare $\hat{F}$ and F^* with a language model³ to assess whether the two are semantically equivalent. In this way we are able to match, e.g., “Alcohol” and “Alcoholic Beverages,” which are semantically equivalent but of different formats. We report accuracy as the metric for this task.

Evaluation of Molecular Profile Completion (MPC). MPC assesses the model’s proficiency in predicting missing flavor molecules.

³In this work we use GPT-3.5-turbo; prompt details are provided in the Appendix B.

Table 1: Classification of food items into macro categories in the dataset.

Categories		
Cereal	Fruit	Essential Oil
Plant	Bakery	Fungus
Seed	Dish	Spice
Flower	Nut and Seed	Beverage
Animal Product	Vegetable	Plant Derivative
Additive	Meat	
Fish and Seafood	Cereal Crop	
Dairy	Herb	

To measure the correctness of predicted missing molecules, we provide the groundtruth answer set M^* . At evaluation time our models predict a set of molecules $\hat{M}$. We thus calculate the F1 score using $f(\hat{M})$ and $f(M^*)$, where f extracts the set of functional groups of the provided set of molecules. Mathematically, the F1 score is calculated as below:

$$F1 = \frac{2 \cdot |f(\hat{M}) \cap f(M^*)|}{|f(\hat{M})| + |f(M^*)|}.$$

By focusing on functional groups, this evaluation protocol leverages the fact that chemicals sharing functional groups tend to exhibit similar properties [28]. In this way we are able to prioritize chemical functionality over strict structural similarity, which ensures predictive relevance and streamline the evaluation process.

4 Evaluated Methods

In this section, we explore selected baseline models (§4.1) and delve into the implementation details of our scientific agent designed for flavor development (§4.2).

4.1 Baseline Methods

We evaluate the following baseline models to establish benchmarks for assessing the capabilities of our proposed scientific agent in the domain of flavor development. Our evaluation is organized into three distinct categories of models.

Foundation LLMs. We evaluate ChatGPT-3.5-turbo,⁴ Gemini-1.5-Pro, and Llama-3-instruct-8B [33] using zero-shot inference. Though being general-purpose and not specifically tailored to food science, these models are renowned for their robust performance across a broad spectrum of natural language processing tasks. The prompts used for zero-shot inference were carefully designed to ensure clarity and relevance to the tasks as shown in Appendix A.

Domain-specific LLMs. We recognize that molecule names and related chemical knowledge can be unfamiliar to general-purpose LLMs. Hence, we also consider models trained on corpora from related domains as baseline methods. These include MolT5, a model pretrained on a vast amount of unlabeled natural language text and molecule strings [6], and BioT5, which is pretrained on structured and unstructured biology corpora for generative tasks involving molecular entities [24]. Since these two models are not instruction-tuned for conversational interaction, we follow Wang et al. [35] to

⁴April 29th 2024 version.

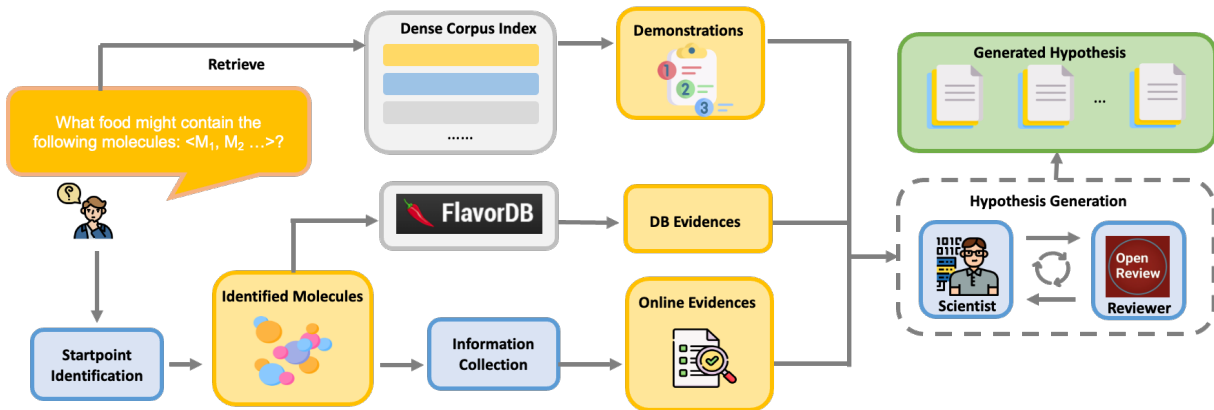


Figure 5: Architecture of the proposed Scientific Agent

constrain the vocabulary set at decoding time and fine-tune these models on the training set for the task.

Both BioT5 and MolT5 are fine-tuned on our train set. We use 4 Nvidia A10G GPUs for our experiments. The models were fine-tuned with a batch size of 16 for the MFP task and 8 for the MPC task, using the AdamW optimizer with a learning rate of 0.00005, $\beta_1 = 0.9$, and $\beta_2 = 0.999$. Each model was trained for 20 epochs, with early stopping applied after 3 epochs based on validation loss.

In-context Learning. We also consider in-context learning techniques for our tasks. However, molecule names may be out-of-context information for LLMs due to limited exposure during pretraining, which may impair agent performance. To enhance reasoning capabilities, we employ a retriever to selectively identify relevant data points from the labeled training set, providing supplementary information. For the implementation of this retriever, we evaluate several approaches: BM25 [26], Sentence Transformer [25], and Dense Passage Retrieval (DPR [17]). Our empirical results indicate that BM25 significantly outperforms dense retrieval methods, achieving a 23.2% accuracy, compared to 13.1% for the Sentence Transformer and 18.0% for DPR in end-to-end performances on the MFC task. This superior performance of BM25 aligns with our expectations, given that dense retrievers are pre-trained and fine-tuned on generic natural language corpora, which may not effectively generalize to retrieve domain-specific data.

For both tasks, the BM25 retriever retrieves the top k most similar demonstrations to the query from a dense corpus index, where each passage is structured in the format: “Food: F' . Molecules: M' ”. These retrieved demonstrations are then presented to the model as demonstrations, which are crucial for generating grounded and evidence-based output.

4.2 Architecture of the Scientific Agent

We propose an advanced hybrid methodology that integrates Retrieval-Augmented Generation (RAG) with online scholarly sources and in-context learning, utilizing pertinent demonstrations. This approach is specifically tailored to surmount the challenges inherent in general-purpose LLMs that lack specialized knowledge in the domain of food science. The architecture of our scientific agent is designed to mirror the investigative processes typical of human

scientists. This alignment not only enhances the agent’s ability to generate scientifically robust hypotheses but also improves the explicability of errors, thereby contributing to its operational transparency and efficacy in flavor science tasks.

Starting Point Identification. Similarly to solving other puzzles, it is important to identify suitable starting points in our flavor profile sourcing task. Consequently, for the MFP task, given a list of flavor molecules, denoted as M , we focus on selecting molecules that are pivotal for our analysis. We calculate the information entropy for each molecule in M based on the frequency of appearances in the train set and select up to 10 molecules with the lowest information entropy to identify an informative subset of molecules. These selected molecules M^* then serve as starting points for subsequent investigations.

Information Collection. One key challenge in employing general-purpose LLMs in specialized fields like food science is their inherent limitation in domain specificity. While these models are trained on massive corpora and varied task datasets that provide a wide range of general knowledge, they often lack the detailed, specific insights needed to tackle complex, domain-focused issues effectively. This shortfall is particularly noticeable in food science, where the models struggle to accurately comprehend or predict the subtle nuances of flavor profiles and the interactions among different compounds. Though in-context learning introduces relevant demonstrations to the agent, it still predominantly relies on the parametric knowledge of LLMs to figure out the relationship between demonstrations and test instances. This inherent limitation can result in outputs that may not sufficiently use in-context information.

To address this gap, Out-of-Domain Evidence is sourced from scholarly articles and internet blogs in addition to the in-context demonstrations. Specifically, for the MFP task, the process begins by iteratively querying each molecule $m \in M^*$. The query is formulated as: “What are the common food sources that could contain m ?” Similarly, for the MPC task, the process begins by querying each food item F , with the corresponding query: “What molecules are present in F ?” We then use the Google Custom Search API to identify academic articles that detail which plants, animals, or other organisms naturally produce these molecules. To ensure the reliability of the information retrieved, the search is confined to

Chemical Name	Collected Info
Elemicin	- Metabolic transformation into reactive metabolites from dietary sources... - Found in nutmeg, containing high levels of myristicin and elemicin... - Present in essential oils (may include eugenol & methyleugenol)... - Extracted from <i>Ocotea pretissima</i>, also found in grass pea... - Studied in tropical fruits, linked to carotenoids...
Tiglic Acid	- Monocarboxylic unsaturated organic acid found in croton oil... - Occurs naturally alongside tropic acid & 2-methylbutyric acid... - Important nutraceutical source (e.g., herbal products, fish oils)... - Often more abundant than angelic acid in most cases... - Found in various plant extracts and natural sources...

Table 2: Examples of chemicals and collected information based on search results that are insightful to MPC task.

scientific repositories such as PubMed⁵ and arXiv.⁶ The scientific agent gathers the sourced articles, storing the information as evidence, as shown in Table 2 and Table 3. To enhance efficiency during inference, the Google Custom Search is executed offline, and the results are stored **locally**.⁷ Consequently, during inference, the system simply retrieves the relevant information from local storage.

Hypothesis Generation. The final phase is hypothesis generation. We realize that it is possible for the collected evidences to sometimes contradict each other, pointing to different answers. Leveraging the Chain-of-Thought (CoT) approach [36], we adopt a role-play based agent framework to process sourced evidences and generate the final hypothesis. Specifically, we initialize two role-based models: the Scientist that proposes three hypotheses and the Reviewer that evaluates the argument quality and selects the best hypothesis⁸. In the MFP task, we retrieve locally stored evidence from FlavorDB related to up to ten key molecules identified during the initial point identification phase. This evidence, along with relevant data points, is then provided to the Scientist model. For the MPC task, as there is no starting point identification process, we retrieve locally evidence related to the food item directly from FlavorDB, and this evidence is provided to the Scientist model. Also, as implemented in the in-context learning baseline, a BM25

⁵<https://pubmed.ncbi.nlm.nih.gov>

⁶<https://arxiv.org>

⁷To facilitate future research, we construct a knowledge base based on collected scholarly articles and release it publicly.

⁸We include prompt examples in Appendix B

Food Item	Collected Info
Dandelion	... Dandelion (<i>Taraxacum officinale</i>) flower ... protein utilizing the principle of... ... The main sesquiterpene compounds in dandelion are sesquiterpene lactones, usually in the form of glycosides, such as sonchuside, cichorioside C... - A single chemical compound in the latex sap of the plant deters cockchafer larvae. ... Dandelions are troublesome weeds that are detested by most... - Due to the interest in the potential pharmacological application of dandelion, the chemical constituents and activities of <i>Taraxacum</i>...
Egg	... This compound is created as sulfur-containing proteins in the egg white react during cooking. The longer an egg is cooked, the greater the... ... Indeed, egg white is a major source of ovostatin, ovomucoid, ovoinhibitor, and cystatin Moreover, some of these molecules (ovoinhibitor)... - The proteins in an egg white are globular proteins, which means that the long protein molecule is twisted and folded and curled up into a more or less spherical... - The eggs of most animals are giant single cells, containing stockpiles of all the materials needed for initial development of the embryo through to the stage at...

Table 3: Example food items and collected information with highlighted chemical segments that are insightful to MFP task.

retriever is used to select the three most similar demonstrations, which are also given to the Scientist model. The Scientist, therefore, receives data input from FOODPUZZLE, the retrieved evidence, and demonstrations, and uses these to generate three hypotheses. The Reviewer model then receives the data input, samples, and the three hypotheses generated by the Scientist. After reviewing the hypotheses, the Reviewer might reject or select the most suitable one. Finally, the scientific model returns this best hypothesis as the final prediction. For budget concern, we focus on benchmarking in-context learning and the scientific agent methods with representative proprietary model (GPT-3.5) and open-source model (LLaMa3-8b-instruct) without loss of generality.



Scientist-agent Prompt

Guidelines:

1. **Generate Hypotheses:** Craft hypotheses using evidence and examples related to key flavor molecules.
2. **Present Clearly:** List three hypotheses, ensuring each is scientifically and culinarily plausible.
3. **Support Claims:** Ensure each hypothesis is well-supported by evidence.

(a) Prompt overview for the Scientist agent.

Reviewer-agent Prompt

Guidelines:

1. **Evaluate:** Judge each hypothesis for scientific validity and culinary plausibility based on evidence and examples.
2. **Select:** Choose the most scientifically rigorous and culinarily relevant hypothesis.
3. **Conclude:** End with a final decision on the best hypothesis.

(b) Prompt overview for the Reviewer agent.

Figure 6: Prompts for the hypothesis generation pipeline.

5 Experiments

In this section, we report and analyze the benchmarking results (§5.1). We randomly select 50 prediction errors and manually inspect them. We present a comprehensive error analysis that outlines key categories of errors as insights for future model refinement (§5.2).

5.1 Main Results

Table 4 presents the combined results for Molecule Food Prediction (MFP) accuracy and Molecule Profile Completion (MPC) F1 scores. As before, the domain-specific LLMs—MolT5 and Bio-T5—lag behind the other models, with MolT5 achieving only 9.8

In the zero-shot setting, the foundation LLMs show a broader range of performance. Gemini1.5 Pro attains an MFP accuracy of 19.0% with an MPC F1 score of 0.340, while Claude-3.5 sonnet reaches the highest MFP accuracy in this group at 23.0% albeit with a slightly lower F1 score of 0.330. GPT-3.5 Turbo and LLaMA3-8B-instruct achieve 12.2% (0.327) and 15.0% (0.292), respectively. These results underscore the inherent challenges of the tasks, suggesting that while general-purpose models can provide reasonable performance, they still lack the fine-tuned domain-specific insights required for food science applications.

The use of in-context learning shows clear benefits. Here, GPT-3.5 Turbo improves its performance to 23.2% MFP accuracy and

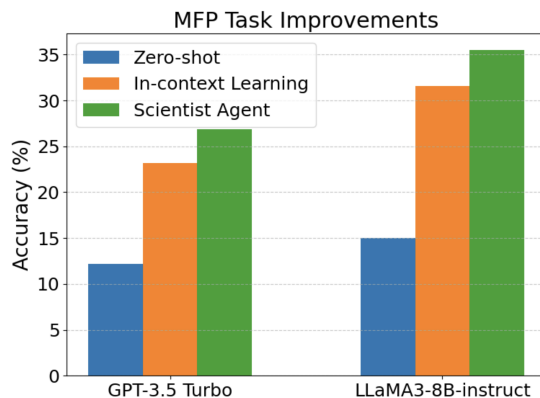


Figure 7: Task performances of different methods on MFP task.

a 0.360 F1 score, and LLaMA3-8B-instruct further raises its MFP accuracy to 31.6% with an F1 score of 0.349. This improvement highlights the importance of leveraging additional contextual information, with our FoodPuzzle dataset proving effective in boosting prediction performance.

Finally, the Scientist Agent approach demonstrates the most robust performance as shown in both Figure 7 and Figure 8. Within this framework, GPT-3.5 Turbo achieves 26.9% MFP accuracy with an MPC F1 score of 0.374, while LLaMA3-8B-instruct stands out with the highest MFP accuracy of 35.5% and matches the top F1 score of 0.374. This superior performance suggests that the agent’s structured strategy—incorporating domain-specific evidence and systematic hypothesis generation—effectively compensates for the limitations observed in both the zero-shot and in-context settings, marking a significant advancement in addressing the challenges posed by food science tasks. The improvements are consistent across both proprietary and open-source model setup.

These findings have important implications for the future development and application of large language models in specialized domains. The demonstrated benefits of incorporating contextual cues and structured reasoning suggest that integrating domain-specific evidence can substantially enhance model performance. This not only bridges the gap between general-purpose and domain-specific models but also paves the way for hybrid approaches that combine the strengths of both.

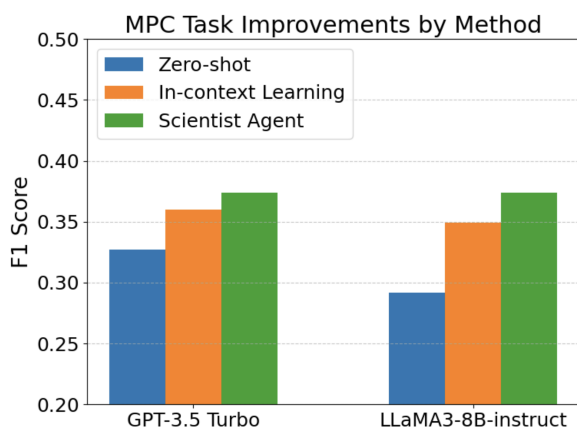
5.2 Error Analysis

Next, we define a taxonomy of error types discerned among evaluated scientific agents, based on an analysis of 50 randomly selected prediction errors. It is possible that a single example may exhibit multiple error types concurrently. To preclude redundancy, only the most salient error per example is accounted for in this analysis.

Inappropriate Initialization of Search Space (32%). When identifying key molecules to understand flavor profiles, the autonomous scientific agents often struggle due to limited domain-specific knowledge. For example, while domain experts can focus on molecules such as “hydrogen sulfide” and “diethyl sulfide” for their significant roles in foods like eggs, LLMs tend to suggest more

Table 4: Combined Results for Model Accuracies in Molecule Food Prediction and F1 Scores for Molecule Profile Completion

Category	Model	MFP accuracy(%)	MPC F1 score
Domain-Specific LLMs	MolT5 [6]	9.8	0.144
	Bio-T5 [24]	16.6	0.278
Zero-shot	Gemini1.5 Pro	19.0	0.340
	Claude-3.5 sonnet	23.0	0.330
	GPT-3.5 Turbo	12.2	0.327
	LLaMA3-8B-instruct	15.0	0.292
In-context Learning	GPT-3.5 Turbo	23.2	0.360
	LLaMA3-8B-instruct	31.6	0.349
Scientist Agent	GPT-3.5 Turbo	26.9	0.374
	LLaMA3-8B-instruct	35.5	0.374

**Figure 8: Task performances of different methods on MPC task.**

generic molecules such as thiamine and betaine. These are common across various foods and can lead the agent to inaccurate, fruit-biased interpretations by narrowing the scope of analysis. This mismatch highlights the importance of incorporating specific and expert-driven insights into the training and inference of LLMs used in flavor science.

Epistemic Hallucination (26%). The phenomenon of hallucination was observed across the evaluated LLMs. During processes intended to enhance model reasoning through chain-of-thought techniques, the output generated by the agent often lacks grounding in factual evidence. For example, when asked to identify flavor molecules for eggs, an LLM might list molecules commonly found in eggs rather than those that could recreate the egg flavor profile. This distinction is crucial: The goal is to identify molecules that can mimic or replicate flavors, not just those that occur naturally in the food item. Effective prompting can mitigate this issue. Clearly specifying that the task involves identifying molecules capable of replicating specific flavors, rather than listing inherent molecules, can direct the LLMs to focus correctly. Additionally, providing explicit examples and context in the prompts can further align the LLM responses with the intended task.

Wrong Interpretation of Online Sources (20%). Another considerable source arises from the misinterpretation of scholarly articles during the process of sourcing evidence online. When models are asked to perform aspect-summarization toward input queries, they exhibit a tendency to “force” answers where direct evidence might be lacking or ambiguous. This compulsion to generate conclusions can lead to erroneous or overly speculative assertions. For instance, consider a scenario where a scholarly article states that ethanethiol is characterized by more roasted and toasted notes, and may exhibit a coffee-like character. LLMs, in its attempt to generate actionable insights, might erroneously infer and assert that “coffee contains Ethanethiol as one of its flavor molecules.” Such a statement is a speculative leap rather than a factually supported conclusion, reflecting the model’s predisposition towards generating responses even in the absence of clear evidence.

6 Future Opportunities

Integrating autonomous flavor scientists and FOODPUZZLE into the R&D pipeline offers significant advancements in flavor science, sensory science, and food product development. In this section we discuss how collaboration with wet lab and sensory scientists to evaluate chemical, biological, and sensory properties of flavor compounds will enhance model accuracy and impact.

Chemical Synthesis and Analysis. Using analytical instruments like LC-MS and GC-MS to detect and quantify flavor molecules in food samples allows for verification of structure and purity. Sensory labs can manage human sensory panels to evaluate synthesized flavors, fine-tuning AI model predictions to align with human perceptions of flavor quality and intensity.

Biological Analysis. Biological laboratories can perform bioassays and in vitro testing to assess the safety and efficacy of flavor compounds. High-throughput screening (HTS) techniques can rapidly test large libraries of flavor molecules, generating extensive datasets to enhance AI predictions.

7 Conclusion

This paper underscores the transformative potential of LLM agents in flavor science, particularly through the development of a scientific agent tailored for flavor profile sourcing. Key contributions



include the formulation of flavor profile sourcing and understanding as a LLM agents task, the creation of the FoodPuzzle dataset from FlavorDB, and the design of a comprehensive scientific agent pipeline that can perform the labor-intensive tasks of flavor sourcing and understanding with enhanced efficiency and precision. Our findings demonstrate that our method outperforms traditional models and provides traceable and reliable predictions. This work not only sets a new benchmark for the application of AI in flavor science but also paves the way for further technological advancements.

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References

- [1] Iz Beltagy, Kyle Lo, and Arman Cohan. 2019. SciBERT: A Pretrained Language Model for Scientific Text. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*. Association for Computational Linguistics, Hong Kong, China, 3615–3620. doi:10.18653/v1/D19-1371
- [2] Sebastian Borgeaud, Arthur Mensch, Jordan Hoffmann, Trevor Cai, Eliza Rutherford, Katie Millican, George Bm Van Den Driessche, Jean-Baptiste Lespiau, Bogdan Damoc, Aidan Clark, Diego De Las Casas, Aurelia Guy, Jacob Menick, Roman Ring, Tom Hennigan, Saffron Huang, Loren Maggiore, Chris Jones, Albin Cassirer, Andy Brock, Michela Paganini, Geoffrey Irving, Oriol Vinyals, Simon Osindero, Karen Simonyan, Jack Rae, Erich Elsen, and Laurent Sifre. 2022. Improving Language Models by Retrieving from Trillions of Tokens. In *Proceedings of the 39th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 162)*, Kamalika Chaudhuri, Stefanie Jegelka, Le Song, Csaba Szepesvari, Gang Niu, and Sivan Sabato (Eds.). PMLR, 2206–2240. <https://proceedings.mlr.press/v162/borgeaud22a.html>
- [3] Andres M Bran, Sam Cox, Oliver Schilter, Carlo Baldassari, Andrew White, and Philippe Schwaller. 2023. Augmenting large language models with chemistry tools. In *NeurIPS 2023 AI for Science Workshop*. <https://openreview.net/forum?id=wdGIL6lx3l>
- [4] Danqi Chen, Adam Fisch, Jason Weston, and Antoine Bordes. 2017. Reading Wikipedia to Answer Open-Domain Questions. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, Vancouver, Canada, 1870–1879. doi:10.18653/v1/P17-1171
- [5] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*. Association for Computational Linguistics, Minneapolis, Minnesota, 4171–4186. doi:10.18653/v1/N19-1423
- [6] Carl Edwards, Tuan Lai, Kevin Ros, Garrett Honke, Kyunghyun Cho, and Heng Ji. 2022. Translation between Molecules and Natural Language. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, Abu Dhabi, United Arab Emirates, 375–413. <https://aclanthology.org/2022.emnlp-main.26>
- [7] Nicki J Engeseth and Marlon Fernando Ac Pangan. 2018. Current context on chocolate flavor development — a review. *Current Opinion in Food Science* 21 (June 2018), 84–91.
- [8] Neelansh Garg, Apuroop Sethupathy, Rudraksh Tuvani, Rakhi Nk, Shubham Dokania, Arvind Iyer, Ayushi Gupta, Shubhra Agrawal, Navjot Singh, Shubham Shukla, et al. 2018. FlavorDB: a database of flavor molecules. *Nucleic acids research* 46, D1 (2018), D1210–D1216.
- [9] Yu Gu, Robert Tinn, Hao Cheng, Michael Lucas, Naoto Usuyama, Xiaodong Liu, Tristan Naumann, Jianfeng Gao, and Hoifung Poon. 2021. Domain-Specific Language Model Pretraining for Biomedical Natural Language Processing. *ACM Trans. Comput. Healthcare* 3, 1, Article 2 (oct 2021), 23 pages. doi:10.1145/3458754
- [10] Kelvin Guu, Kenton Lee, Zora Tung, Panupong Pasupat, and Mingwei Chang. 2020. Retrieval Augmented Language Model Pre-Training. In *Proceedings of the 37th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 119)*, Hal Daumé III and Aarti Singh (Eds.). PMLR, 3929–3938. <https://proceedings.mlr.press/v119/guu20a.html>
- [11] Leah M Hamilton and Jacob Lahne. 2020. Fast and automated sensory analysis: Using natural language processing for descriptive lexicon development. *Food Qual. Prefer.* 83 (July 2020), 103926.
- [12] Thomas Hofmann, Dietmar Krautwurst, and Peter Schieberle. 2018. Current Status and Future Perspectives in Flavor Research: Highlights of the 11th Wartburg Symposium on Flavor Chemistry & Biology. *J. Agric. Food Chem.* 66, 10 (March 2018), 2197–2203.
- [13] Sirui Hong, Yizhang Lin, Bangbang Liu, Binhao Wu, Danyang Li, Jiaqi Chen, Jiayi Zhang, Jinlin Wang, Lingyao Zhang, Mingchen Zhuge, Taicheng Guo, Tuo Zhou, Wei Tao, Wenyi Wang, Xiangru Tang, Xiangtao Lu, Xinbing Liang, Yaying Fei, Yuheng Cheng, Zongze Xu, Chenglin Wu, Li Zhang, Min Yang, and Xiwu Zheng. 2024. Data Interpreter: An LLM Agent For Data Science. *ArXiv abs/2402.18679* (2024). <https://api.semanticscholar.org/CorpusID:268063292>
- [14] Sameera Horawalavithana, Ellyn Ayton, Shivam Sharma, Scott Howland, Megha Subramanian, Scott Vasquez, Robin Cosbey, Maria Glenski, and Svitlana Volkova. 2022. Foundation Models of Scientific Knowledge for Chemistry: Opportunities, Challenges and Lessons Learned. In *Proceedings of BigScience Episode #5 – Workshop on Challenges & Perspectives in Creating Large Language Models*. Association for Computational Linguistics, virtual+Dublin, 160–172. doi:10.18653/v1/2022.bigscience-1.12
- [15] Tenghao Huang, Kinjal Basu, Ibrahim Abdelaziz, Pavan Kapanipathi, Jonathan May, and Muhao Chen. 2025. R2D2: Remembering, Reflecting and Dynamic Decision Making for Web Agents. *arXiv preprint arXiv:2501.12485* (2025).
- [16] Tenghao Huang, Dongwon Jung, Vaibhav Kumar, Mohammad Kachuee, Xiang Li, Puyang Xu, and Muhao Chen. 2024. Planning and Editing What You Retrieve for Enhanced Tool Learning. In *Findings of the Association for Computational Linguistics: NAACL 2024*, 975–988.
- [17] Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. 2020. Dense Passage Retrieval for Open-Domain Question Answering. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics, Online, 6769–6781. doi:10.18653/v1/2020.emnlp-main.550
- [18] Jinhyuk Lee, Wonjin Yoon, Sungdong Kim, Donghyeon Kim, Sunkyu Kim, Chan Ho So, and Jaewoo Kang. 2019. BioBERT: a pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics* 36, 4 (09 2019), 1234–1240. doi:10.1093/bioinformatics/btz682 [arXiv:https://academic.oup.com/bioinformatics/article-pdf/36/4/1234/48983216/bioinformatics_36_4_1234.pdf](https://academic.oup.com/bioinformatics/article-pdf/36/4/1234/48983216/bioinformatics_36_4_1234.pdf)
- [19] Kenton Lee, Ming-Wei Chang, and Kristina Toutanova. 2019. Latent Retrieval for Weakly Supervised Open Domain Question Answering. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics, Florence, Italy, 6086–6096. doi:10.18653/v1/P19-1612
- [20] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Kuttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. In *Advances in Neural Information Processing Systems*, H. Larochelle, M. Ranzato, R. Hadsell, M.F. Balcan, and H. Lin (Eds.), Vol. 33. Curran Associates, Inc., 9459–9474. https://proceedings.neurips.cc/paper_files/paper/2020/file/6b49320205f780e1bc26945df7481e5-Paper.pdf
- [21] Jieyu Lu and Yingkai Zhang. 2022. Unified Deep Learning Model for Multitask Reaction Predictions with Explanation. *Journal of Chemical Information and Modeling* (2022).
- [22] Odhran O'Donoghue, Aleksandar Shtedritski, John Ginger, Ralph Abboud, Ali Ghareeb, and Samuel Rodrigues. 2023. BioPlanner: Automatic Evaluation of LLMs on Protocol Planning in Biology. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2676–2694.
- [23] Jayvadan Patel and Anita Patel. 2019. Flavor Manufacturing and Selection Criteria for Functional Food and Nutraceuticals Industries. In *Flavor Development for Functional Foods and Nutraceuticals*. CRC Press, 39–72.
- [24] Qizhi Pei, Wei Zhang, Jinhua Zhu, Kehan Wu, Kaiyuan Gao, Lijun Wu, Yingce Xia, and Rui Yan. 2023. BioT5: Enriching Cross-modal Integration in Biology with Chemical Knowledge and Natural Language Associations. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, Houda Bouamor, Juan Pino, and Kalika Bali (Eds.). Association for Computational Linguistics, Singapore, 1102–1123. doi:10.18653/v1/2023.emnlp-main.70
- [25] Nils Reimers and Iryna Gurevych. 2019. Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics. <http://arxiv.org/abs/1908.10084>
- [26] S. E. Robertson and S. Walker. 1994. Some simple effective approximations to the 2-Poisson model for probabilistic weighted retrieval. In *Proceedings of the 17th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval* (Dublin, Ireland) (SIGIR '94). Springer-Verlag, Berlin, Heidelberg, 232–241.
- [27] Tatsuya Sagawa and Ryosuke Kojima. 2023. ReactionT5: a large-scale pre-trained model towards application of limited reaction data. *arXiv:2311.06708 [physics.chem-ph]*

- [28] Elena S. Salmina, Norbert Haider, and Igor V. Tetko. 2016. Extended Functional Groups (EFG): An Efficient Set for Chemical Characterization and Structure-Activity Relationship Studies of Chemical Compounds. *Molecules* 21, 1 (2016). doi:10.3390/molecules21010001
- [29] K. Singhal, Shekoofeh Azizi, Tao Tu, Said Mahdavi, Jason Wei, Hyung Won Chung, Nathan Scales, Ajay Kumar Tanwani, Heather J. Cole-Lewis, Stephen J. Pfohl, P A Payne, Martin G. Seneviratne, Paul Gamble, Chris Kelly, Nathaneal Scharli, Aakanksha Chowdhery, P. A. Mansfield, Blaise Agera y Arcas, Dale R. Webster, Greg S. Corrado, Yossi Matias, Katherine Hui-Ling Chou, Juraj Gottweis, Nenad Tomaev, Yun Liu, Alvin Rajkomar, Jolle K. Barral, Christopher Semturs, Alan Karthikesalingam, and Vivek Natarajan. 2022. Large language models encode clinical knowledge. *Nature* 620 (2022), 172 – 180. <https://api.semanticscholar.org/CorpusID:255124952>
- [30] Yu Song, Santiago Miret, and Bang Liu. 2023. MatSci-NLP: Evaluating Scientific Language Models on Materials Science Language Tasks Using Text-to-Schema Modeling. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Anna Rogers, Jordan Boyd-Graber, and Naoaki Okazaki (Eds.). Association for Computational Linguistics, Toronto, Canada, 3621–3639. doi:10.18653/v1/2023.acl-long.201
- [31] Alexander Spangher, Tenghao Huang, Philippe Laban, and Nanyun Peng. 2025. Creative Planning with Language Models: Practice, Evaluation and Applications. In *Proceedings of the 2025 Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 5: Tutorial Abstracts)*, Maria Lomeli, Swabha Swayamdipta, and Rui Zhang (Eds.). Association for Computational Linguistics, Albuquerque, New Mexico, 1–9. <https://aclanthology.org/2025.naacl-tutorial.1/>
- [32] Ross Taylor, Marcin Kardas, Guillem Cucurull, Thomas Scialom, Anthony Hartshorn, Elvis Saravia, Andrew Poulton, Viktor Kerkez, and Robert Stojnic. 2022. Galactica: A Large Language Model for Science. arXiv:2211.09085 [cs.CL]
- [33] Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, Dan Bikel, Lukas Blecher, Cristian Canton Ferrer, Moya Chen, Guillem Cucurull, David Esiobu, Jude Fernandes, Jeremy Fu, Wenyin Fu, Brian Fuller, Cynthia Gao, Vedanuj Goswami, Naman Goyal, Anthony Hartshorn, Saghar Hosseini, Rui Hou, Hakan Inan, Marcin Kardas, Viktor Kerkez, Madian Khabsa, Isabel Kloumann, Artem Korenev, Punit Singh Koura, Marie-Anne Lachaux, Thibaut Lavril, Jenya Lee, Diana Liskovich, Yinghai Lu, Yuning Mao, Xavier Martinet, Todor Mihaylov, Pushkar Mishra, Igor Molybog, Yixin Nie, Andrew Poulton, Jeremy Reizenstein, Rashi Rungta, Kalyan Saladi, Alan Schelten, Ruan Silva, Eric Michael Smith, Ranjan Subramanian, Xiaoqing Ellen Tan, Binh Tang, Ross Taylor, Adina Williams, Jian Xiang Kuan, Puxin Xu, Zheng Yan, Iliyan Zarov, Yuchen Zhang, Angela Fan, Melanie Kambadur, Sharan Narang, Aurelien Rodriguez, Robert Stojnic, Sergey Edunov, and Thomas Scialom. 2023. Llama 2: Open Foundation and Fine-Tuned Chat Models. arXiv:2307.09288 [cs.CL]
- [34] Jesse Vig, Ali Madani, Lav R. Varshney, Caiming Xiong, richard socher, and Nazneen Rajani. 2021. {BERT}ology Meets Biology: Interpreting Attention in Protein Language Models. In *International Conference on Learning Representations*. <https://openreview.net/forum?id=YWtLZvLmud7>
- [35] Fei Wang, Zhewei Xu, Pedro Szekely, and Muhao Chen. 2022. Robust (Controlled) Table-to-Text Generation with Structure-Aware Equivariance Learning. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Association for Computational Linguistics, Seattle, United States, 5037–5048. doi:10.18653/v1/2022.naacl-main.371
- [36] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, brian ichter, Fei Xia, Ed Chi, Quoc V Le, and Denny Zhou. 2022. Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. In *Advances in Neural Information Processing Systems*, S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh (Eds.), Vol. 35. Curran Associates, Inc., 24824–24837. https://proceedings.neurips.cc/paper_files/paper/2022/file/9d5609613524ecf4f15af0f7b31abca4-Paper-Conference.pdf
- [37] Jiarui Zhang. 2024. Guided Profile Generation Improves Personalization with Large Language Models. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen (Eds.). Association for Computational Linguistics, Miami, Florida, USA, 4005–4016. doi:10.18653/v1/2024.findings-emnlp.231

A Prompt Details

Figure 9, and Figure 10 show the prompt details that we interact with the Foundation LLMs. Figure 11, Figure 12 show prompts used for Autonomous Scientific Agent.

**Molecular Food Prediction Prompt for Foundation LLMs with ICL**

You are a food scientist AI. Your task is to predict the food products based on a given list of flavor molecules. Provide concise and accurate predictions. Ensure that your response is strictly in the JSON format {'food': 'food_name'}. Only provide the name of one food product that is most likely to contain the given flavor molecules. You will be also given similar examples in the prompts. Use them as reference to make your predictions.

Example 1: {*ICL example 1*}

Example 2: {*ICL example 2*}

Example 3: {*ICL example 3*}

Now, what food products might contain the following flavor molecules: {*Molecule List*}? Please return the result strictly in the format {'food': 'food_name'}.

(a) Molecular Food Prediction Prompt for foundation LLMs with In-Context Learning

Molecular Profile Completion Prompt for Foundation LLMs with ICL

You are a powerful food scientist specializing in predicting missing molecules from a list of incomplete flavor molecules and food sources. You will be given a food source, a list of incomplete flavor molecules, and the number of missing molecules, and you need to predict the missing molecules that might be missing from the given molecule list. Consider common molecular profiles and typical combinations in the food source. You will be also given similar examples in the prompts. Use them as reference to predict the missing molecules.

Example 1: {*ICL example 1*}

Example 2: {*ICL example 2*}

Example 3: {*ICL example 3*}

Based on the examples provided, identify the missing molecules in the following case:

Food: {*Food*}

Incomplete molecules: {*Molecule List*}

Number of missing molecules: {*N*}

Provide the answer in the following format: [Reasoning]

Missing molecules: ['mol1', 'mol2', 'mol3']

(b) Molecular Profile Completion Prompt for foundation LLMs with In-Context Learning

Figure 10: Foundation LLMs with In-Context Learning Prompts

Molecular Food Prediction System Prompt for Autonomous Scientific Agent

You are a part of a sophisticated multi-agent system designed to predict food sources from specific flavor molecules. This system incorporates several stages of analysis, including evidence gathering, hypothesis generation, and critical evaluation. Each agent in the system plays a vital role:

- **Retriever**: Fetches relevant examples and data about the given query molecules.
- **Analyst**: Identifies key molecules that are indicative of specific food types based on the retrieved data.
- **Engineer**: Receives snippets of scientific papers related to the indicative molecules.
- **Scientist**: Uses the snippets provided by the Engineer to propose three scientifically sound food hypotheses, emphasizing scientific validity and culinary plausibility.
- **Reviewer**: Evaluates each hypothesis for its scientific and culinary merits. Based on the assessment, the Reviewer selects the best hypothesis that best meets the required standards.

Your collaboration is iterative, with each role building upon the work of the previous. For example, if a combination like ['Vanillin', 'Limonene', 'Capsaicin'] is given, the Retriever would fetch similar case studies, the Analyst identifies key molecules, the Engineer summarizes relevant scientific findings, the Scientist formulates three potential food hypotheses, and the Reviewer assesses these hypotheses, selecting the most plausible one.

Stay focused on your individual roles, collaborate effectively, and aim to derive a well-informed and valid hypothesis.

Objective:

Leverage collaborative insights and detailed scientific evidence to predict the most plausible food sources corresponding to the given combination of flavor molecules.

Let's work together to achieve this sophisticated task.

Figure 11: Molecular Food Prediction System Prompt for Autonomous Scientific Agent

Molecular Food Prediction *Scientist* Prompt for Autonomous Scientific Agent

You are the Scientist. Your task is to propose three food hypotheses based on the evidences from the Engineer, and the examples from Retriever. The evidences and examples are derived from scientific literature related to the key flavor molecules identified earlier in the process. Here's how you should approach your responsibilities:

- **Hypothesis Generation**: Utilize the detailed summaries provided by the Engineer, which include crucial scientific evidence about the culinary implications of the key molecules. From this information, craft three potential hypotheses about what foods these molecules could be associated with. Each hypothesis must include a valid food name and should be scientifically grounded and culinarily plausible.
- **Clear Presentation**: Clearly state your three proposed hypotheses. Ensure to strictly follow the format as follows to ensure clarity and ease of evaluation by the Reviewer:
 Hypothesis 1. [Valid, commonly recognized food name]
 [Explanation]

 Hypothesis 2. [Valid, commonly recognized food name]
 [Explanation]

 Hypothesis 3. [Valid, commonly recognized food name]
 [Explanation]

Your ability to derive well-informed, valid hypotheses from the provided evidence is critical. These hypotheses will be rigorously evaluated for their scientific coherence and culinary plausibility by the Reviewer, who will either select the best hypothesis or provide feedback for refinement.

Based on the following evidence and retrieved examples related to the query molecules *{Molecule List}*, propose three scientifically sound and culinarily plausible food hypotheses. Each hypothesis should be distinct and directly supported by the evidence provided.

Evidence: *{Evidence}*
 Examples from Retriever: *{ICL examples}*

Figure 12: Molecular Food Prediction *Scientist* Prompt for Autonomous Scientific Agent

Molecular Food Prediction Prompt for Foundation LLMs

You are a food scientist AI. Your task is to predict the food products based on a given list of flavor molecules. Provide concise and accurate predictions. Ensure that your response is strictly in the JSON format `{'food': 'food_name'}`. Only provide the name of one food product that is most likely to contain the given flavor molecules.

What food products might contain the following flavor molecules: *{Molecule List}*? Please return the result strictly in the format `{'food': 'food_name'}`.

(a) Molecular Food Prediction Prompt for Foundation LLMs

Molecular Profile Completion Prompt for Foundation LLMs

You are a powerful food scientist specializing in predicting missing molecules from a list of incomplete flavor molecules and food sources. You will be given a food source, a list of incomplete flavor molecules, and the number of missing molecules, and you need to predict the missing molecules that might be missing from the given molecule list. Consider common molecular profiles and typical combinations in the food source.

Food: *{Food}*
 Incomplete molecules: *{Molecule List}*
 Number of missing molecules: *{N}*
 Provide the answer in the following format: [Reasoning]
 Missing molecules: ['mol1', 'mol2', 'mol3']

(b) Molecular Profile Completion Prompt for Foundation LLMs

Figure 9: Foundation LLMs Prompts